



Applications: Series

Series have many applications in everyday life.

Determining Whether a Series Is Arithmetic, Geometric, or Neither

Before you use series to solve problems, you should know how to tell whether a series is arithmetic, geometric, or neither.

A geometric series is a series for which the terms are the elements of a geometric sequence. An arithmetic series is a series for which the terms are the elements of an arithmetic sequence.

EXAMPLE 1

Determine whether the series is arithmetic, geometric, or neither.

A $4 + 12 + 36 + 108 + \dots$

SOLUTION

Test to see if it is arithmetic.

$$a_2 - a_1 = 12 - 4 = 8 \quad a_3 - a_2 = 36 - 12 = 24$$

The differences are not the same, so test to see if it is geometric.

$$\frac{a_2}{a_1} = \frac{12}{4} = 3 \quad \frac{a_3}{a_2} = \frac{36}{12} = 3 \quad \frac{a_4}{a_3} = \frac{108}{36} = 3$$

The ratios are the same, so the series is geometric.

B $1 + 2 + 6 + 24 + \dots$

SOLUTION

Test to see if it is arithmetic.

$$a_2 - a_1 = 2 - 1 = 1 \quad a_3 - a_2 = 6 - 2 = 4$$

The differences are not the same, so test to see if it is geometric.

$$\frac{a_2}{a_1} = \frac{2}{1} = 2 \quad \frac{a_3}{a_2} = \frac{6}{2} = 3$$

The ratios are not the same, so the series is neither arithmetic nor geometric.

C $480 + 360 + 240 + 120 + \dots$

SOLUTION

Test to see if it is arithmetic.

$$a_2 - a_1 = 360 - 480 = -120$$

$$a_3 - a_2 = 240 - 360 = -120$$

$$a_4 - a_3 = 120 - 240 = -120$$

The differences are the same, so the series is arithmetic.

D $-8 + 4 - 2 + 1 + \dots$

SOLUTION

Test to see if it is arithmetic.

$$a_2 - a_1 = 4 - (-8) = 12 \quad a_3 - a_2 = -2 - 4 = -6$$

The differences are not the same, so test to see if it is geometric.

$$\frac{a_2}{a_1} = \frac{4}{-8} = -\frac{1}{2} \quad \frac{a_3}{a_2} = \frac{-2}{4} = -\frac{1}{2} \quad \frac{a_4}{a_3} = \frac{1}{-2} = -\frac{1}{2}$$

The ratios are the same, so the series is geometric. ■

Application: Money

EXAMPLE 2

Brian and Kayla try different methods of saving money. Determine the total amount of money each will have after 10 days.

A Brian saves 51¢ on Day 1, 52¢ on Day 2, 53¢ on Day 3, . . .

SOLUTION

The total amount of money Brian saves after 10 days, given 51¢ on Day 1, 52¢ on Day 2, 53¢ on Day 3, . . . , is an arithmetic series $51 + 52 + 53 + 54 + \dots$, where $n = 10$, $a_1 = 51$, and $d = 1$.

$$\begin{aligned}S_n &= \frac{n}{2} (2a_1 + (n-1)d) \\S_{10} &= \frac{10}{2} (2 \cdot 51 + (10-1) \cdot 1) \\S_{10} &= 5(102 + 9 \cdot 1) \\S_{10} &= 5 \cdot 111 = 555\end{aligned}$$

The total amount of money Brian saves after 10 days is \$5.55.

B Kayla saves 1¢ on Day 1, 2¢ on Day 2, 4¢ on Day 3, . . .

SOLUTION

The total amount of money Kayla saves after 10 days, given 1¢ on Day 1, 2¢ on Day 2, 4¢ on Day 3, . . . , is a geometric series $1 + 2 + 4 + 8 + \dots$, where $n = 10$, $a_1 = 1$, and $r = 2$.

$$\begin{aligned}S_n &= \frac{a_1(1-r^n)}{1-r} \\S_{10} &= \frac{1(1-2^{10})}{1-2} \\S_{10} &= \frac{1 \cdot (-1023)}{-1} \\S_{10} &= 1023\end{aligned}$$

The total amount of money Kayla saves after 10 days is \$10.23. Kayla saves more money with her method than Brian saves with his. ■

Application: Seating Capacity

EXAMPLE 3

Suppose an auditorium has 30 rows of seats. There are 30 seats in the first row, 32 in the second row, 34 in the third row, and so on. How many seats are in all 30 rows?

SOLUTION

The total number of seats in the 30 rows form an arithmetic series with $n = 30$, $a_1 = 30$, and $d = 2$.

Step 1 Find the 30th term.

$$\begin{aligned}a_n &= a_1 + (n - 1)d \\a_{30} &= 30 + (30 - 1) \cdot 2 \\a_{30} &= 30 + 29 \cdot 2 \\a_{30} &= 30 + 58 \\a_{30} &= 88\end{aligned}$$

Step 2 Find the sum of the series.

$$\begin{aligned}S_n &= \frac{n}{2}(a_1 + a_n) \\S_{30} &= \frac{30}{2}(30 + 88) \\S_{30} &= 15 \cdot 118 \\S_{30} &= 1770\end{aligned}$$

There are 1770 seats in the auditorium. ■

Application: Bouncing Distance

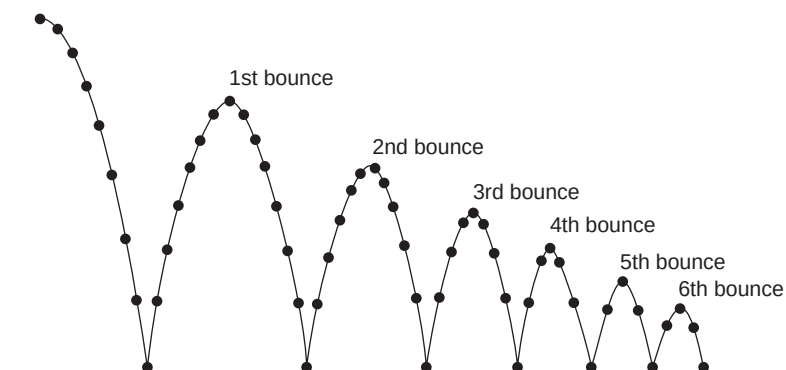
EXAMPLE 4

Silas drops a ball from a height of 25 ft. Each time it drops h feet, it rebounds to a height of $0.8h$ feet. How much vertical distance does the ball travel from the time he drops it until it reaches the peak after its 6th bounce?

SOLUTION

The ball travels 25 ft on the first drop. On each bounce after that, it goes up to its peak, and then back down the same distance, so it travels $2 \cdot 25 \cdot 0.8^k$ feet for $k = 1$ to 5. The ball then goes up after the 6th bounce.

$$25 \cdot 0.8^6 = 6.5536$$



The five bounces, up and down, form the geometric series

$$2 \cdot 25 \cdot 0.8^1 + 2 \cdot 25 \cdot 0.8^2 + 2 \cdot 25 \cdot 0.8^3 + 2 \cdot 25 \cdot 0.8^4 + 2 \cdot 25 \cdot 0.8^5$$

where $n = 5$, $a_1 = 2 \cdot 25 \cdot 0.8^1 \approx 40$, and $r = 0.8$.

Substitute the values of n , a_1 , and r into the formula,

$$\begin{aligned} S_n &= \frac{a_1(1 - r^n)}{1 - r} \\ S_5 &= \frac{40(1 - 0.8^5)}{1 - 0.8} \\ S_5 &= \frac{40(0.67232)}{0.2} \\ S_5 &= \frac{26.8928}{0.2} \\ S_5 &= 134.464 \end{aligned}$$

The total vertical distance the ball travels is

$$25 + 134.464 + 6.5536 = 166.0176.$$

The ball travels a vertical distance of about 166 ft. ■

Application: Salary with Raises

EXAMPLE 5

Maya starts working for Titanium Works Company with a salary of \$30,000. If she gets a 5% raise at the end of each year, what will Maya's total earnings be after 30 years on the job?

SOLUTION

The salaries form a geometric sequence with $a_1 = 30,000$ and $r = 1.05$.

Year	Raise from Previous Year	Salary
1		30,000
2	$30,000 \cdot 0.05 = 1500$	$30,000 \cdot 1.05^1 = 31,500$
3	$31,500 \cdot 0.05 = 1575$	$30,000 \cdot 1.05^2 = 33,075$
4	$33,075 \cdot 0.05 = 1653.75$	$30,000 \cdot 1.05^3 = 34,728.75$

To find Maya's total earnings, find the sum of the first 30 terms ($n = 30$) of the geometric series.

$$\begin{aligned}S_n &= \frac{a_1(1 - r^n)}{1 - r} \\S_{30} &= \frac{30,000(1 - 1.05^{30})}{1 - 1.05} \\S_{30} &\approx \frac{30,000(-3.32194)}{-0.05} \\S_{30} &\approx \frac{-99,658.27125}{-0.05} \\S_{30} &\approx 1,993,164.425\end{aligned}$$

Maya's total earnings for 30 years will be about \$1,993,164. ■

